

SEBASTIAN MONZÓN

Full-Stack Engineer & Researcher

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RESEARCH INTERESTS

Human-Computer Interaction, Cognitive AI, Affective Computing, Music Technology

EDUCATION

B.S. in Computer Science

University of Massachusetts Amherst, [CGPA 3.7/4.0](#)

Amherst, MA
09/2019 - 05/2023

- Honors Thesis from the Commonwealth Honors College
- Business Minor

AWARDS AND HONORS

NSF Fellow (CSGrad4US)

\$159,000 covering 3 years over a 5-year fellowship period starting Fall 2026.

Scholarships

UMass Amherst Dean's Award
Melvin Howard Scholarship
Ruth M. White Scholarship

HackUMass IX Hackathon Awards

Won a finalist award and "Best use of Google Cloud" for a Spotify song voting queue app.

PUBLICATIONS IN-PREPARATION

An AI-Driven Acoustic Symphony: Optimized Hydrogel Dewatering Leveraging a Diffusion-model for Music Spectrogram Inpainting

Shuvo, I., Monzón, S., Smith, G., Flores, R., Diaz-Marin, C., Gu, R., Nagle, S., Paradiso, J., Boriskina, V., & Boriskina, S.
Designed and tested an AI framework for hydrogel dewatering optimization using music generated via Stable Diffusion and soft-prompt optimization.

A Wearable Waterproof System for Long-Range Health Monitoring and Intelligent Distress Signaling

Shuvo, I., Monzón, S., & Boriskina, S.
Modeled, fabricated, and coded a waterproof device with long-range radio (LoRa), IMU, and heart rate sensor for real-time machine learning interpretation of human movement and biomedical data for swimmers.

Musically Interpreting a Simple Generative Transformer

Cosme-Clifford, N., Symons, J., Kapoor, K., Monzón, S., & Wm. White, C.
Explored the aptitudes and failures of a GPT-style model trained on choral-style hymn MIDI files. Encoded chords into Unicode for GPT interpretation and analyzed its learned latent space for implications of AI on music.

MAGE: Motion-to-Audio Generative autoEncoder for Expressive Real-Time Hand-Gesture-Conditioned Sound Synthesis

Monzón, S.
Independently invented a multi-modal variational auto-encoder (MMVAE) using a Realtime Audio Variational autoEncoder (RAVE) variant and a Gated Recurrent Unit (GRU) for hand-gesture-to-audio synthesis. Uses high FPS hand positioning to encode hand movement into real-time conga drum sounds.

CONFERENCES & PRESENTATIONS

Symphony of Dewatering Water Harvesting Hydrogels using AI-Generated Music

2025 Material Research Society Fall Meeting, Boston

12/2025

Presenting an extension of my original framework using soft-token prompt optimization and similar Stable Diffusion methods.

Music Generation with Stable Diffusion Inpainting

Dept. of Multifunctional Metamaterials (META) at MIT

04/2025

Presented research on the AI framework I developed to improve hydrogel dewatering using Stable Diffusion "inpainting" methods.

ADDITIONAL RESEARCH EXPERIENCE

Music-Emotion Study

Amherst, MA

Department of Music, UMass Amherst

08/2025

- Composing 2 pop musical pieces with fixed chord progressions to measure physiological states with control across other music genres.
- Collaborated to determine 4 measurable response metrics for musical control and consistency to reduce data variance.

Undergraduate Honors Thesis

Amherst, MA

Commonwealth Honors College, UMass Amherst

04/2023

"Towards Generative Adversarial Audio Synthesis via Latent Interpolation Between Desired Timbral Qualities"

- Researched state-of-the-art AI audio generation models including WaveGAN, WaveNet, and NSynth.
- Proposed an AI model to generate musical notes based on sound descriptors using a GAN model.
- Designed and built a multi-class classification network to categorize sounds by their instruments with 76% accuracy.
- Wrote a peer-reviewed 24-page manuscript.

INDUSTRY EXPERIENCE

Software Engineer

Remote

Cisco Meraki

06/2023 - Present

- Collaborated closely with UX designers and project managers to design, develop, and deploy scalable networking solutions for the Meraki Dashboard.
- Technical Lead for the "Power Settings" page for the Meraki Dashboard to distribute power-saving configuration options to Meraki APs.
- Resolved several time-sensitive, million dollar issues for clients across the globe by debugging Meraki's 4 million line codebase.
- Performed several interviews and onboarded software engineers in India and China.
- Designed an LLM for retrieving Slack and Confluence documents to help developers access information during the 1-week Cisco hackathon.

Startup CTO

Boston, MA

Beamshyft

2025 - Present

beamshyft.com

Low-cost housing supplies shipped directly from manufacturers

- Designed, developed, and launched a professional, dynamic website.
- Improved project clarity by outlining phases in a statement of work for 4 key development phases, enhancing partner understanding.
- Adapted development phases to meet business needs.

Software Engineer Intern

Remote

Cisco Meraki

2023

- Collaborated the UX team to develop the updated UI for the Radio Settings page using Cisco Magnetic design principles.
- Diagnosed high-priority issues to prevent dashboard outages for customers with large networks.

UNDERGRADUATE COURSES

Artificial Intelligence

Intro to Machine Learning

Reinforcement Learning (M.L. colloquium)

Uncertainty, Risk, and Decision Making

Software Entrepreneurship

Intro to Music Theory

ACTIVITIES

Swim-team captain, robotics club (computer vision), St. Jude Children's Research Hospital fundraising, TKE recruitment chairman

SKILLS

AI & Machine Learning:

PyTorch, Tensorflow, MediaPipe, Multimodal VAE, VRNN, Audio/Signal Processing, CLIP, Stable Diffusion, RAVE/BRAVE

Programming & Systems:

Python, Matplotlib, Librosa, Pandas, Embedded C++, CUDA, MATLAB, React/Next.js, Ruby on Rails, SQL

Design & Fabrication: CAD (Fusion 360), PCB Design (KiCAD), 3D Printing

Creative Practice: Music Production (Ableton Live), Sound Design, Adobe

Research & Interpersonal:

Problem-Solving & Critical Thinking, Creativity & Innovation (Musician, Inventor, Builder), Collaboration & Communication (Research Partnerships, Recruitment Chairman), Leadership & Initiative (Swim Team Captain, Tech Lead), Adaptability & Resilience, Industrious Work Ethic

LANGUAGES

English (Native)

Spanish (Proficient)